Riphah International University Sahiwal

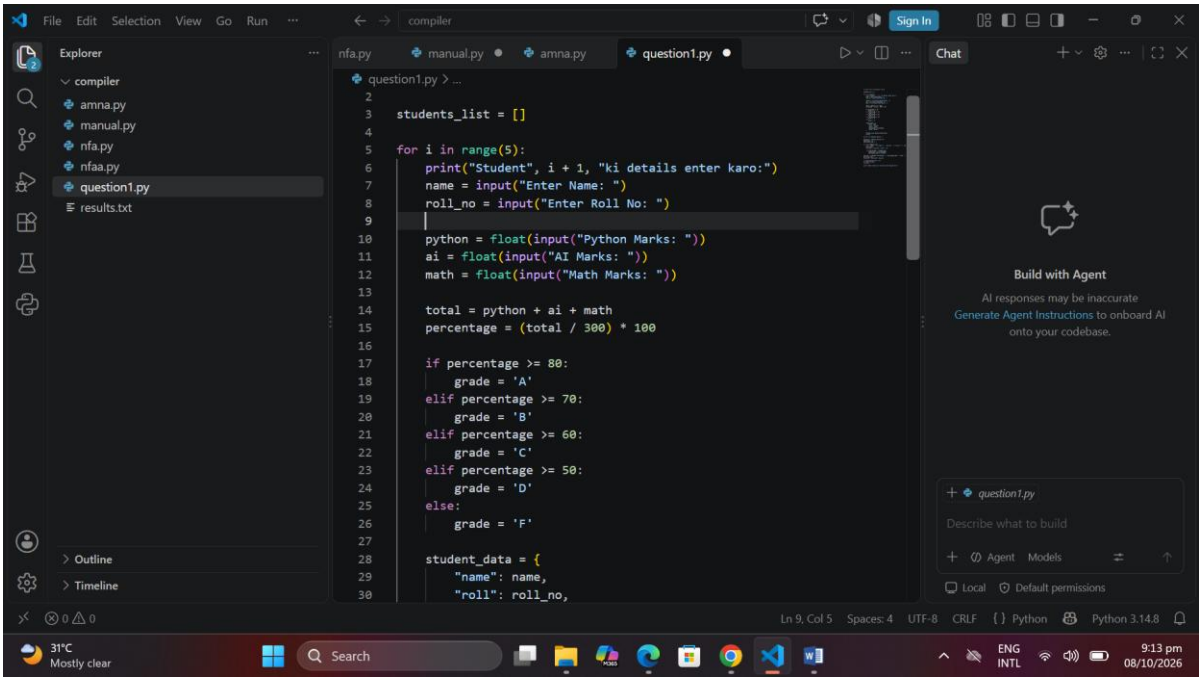
DEPARTMENT OF Computing And Innovation

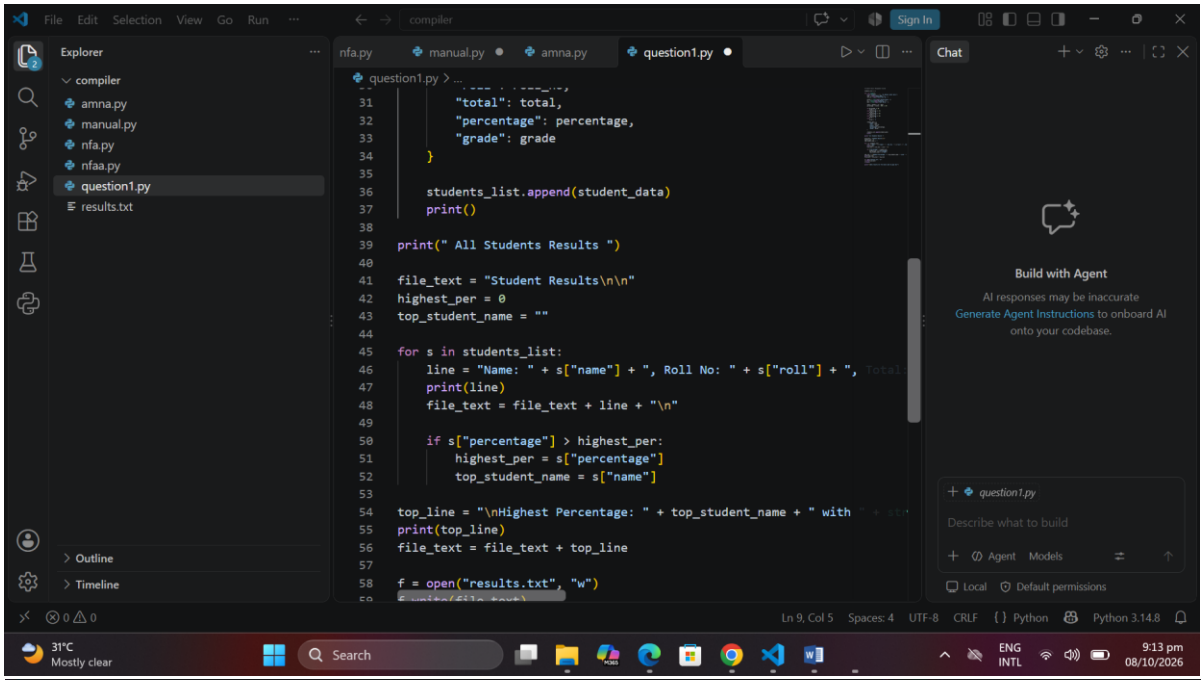
Assignment

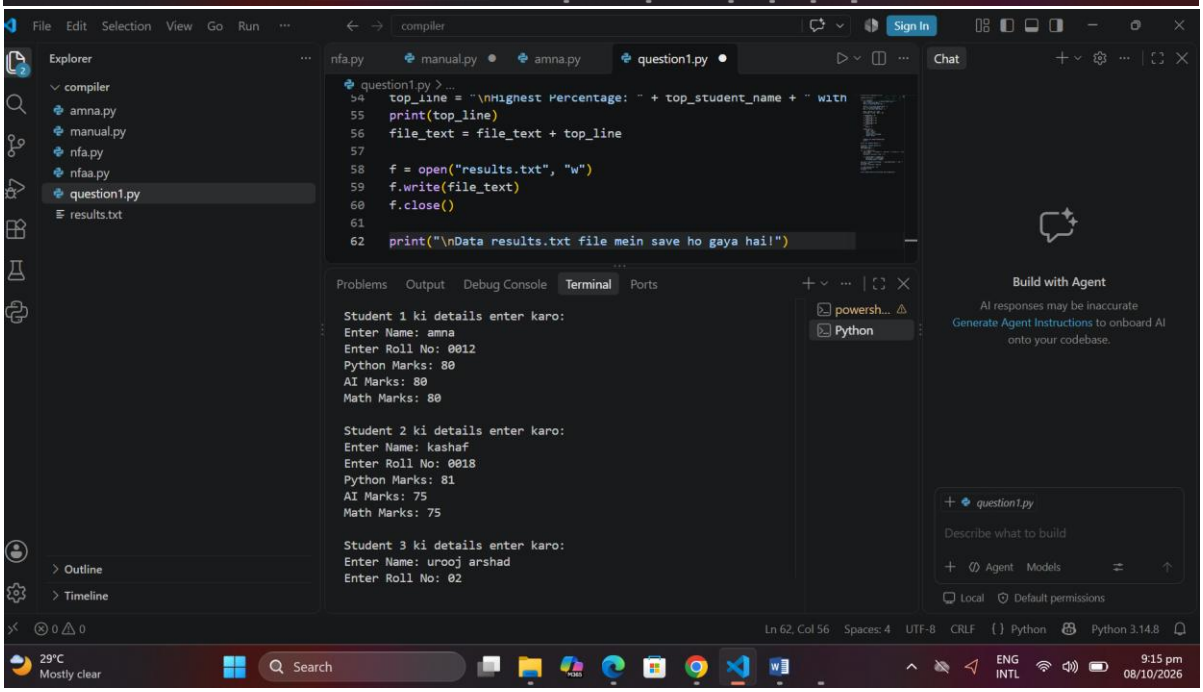
Course Title:	**Lab of Fundamentals of AI** _____	**Cms no:**	___0012_____
Name:	**Amna Mahnoor** _____	**Semester / Section:**	___SP 24 CS&SE_____
Assignment No.:	___01_____	**Date:**	___08/10/2026_____

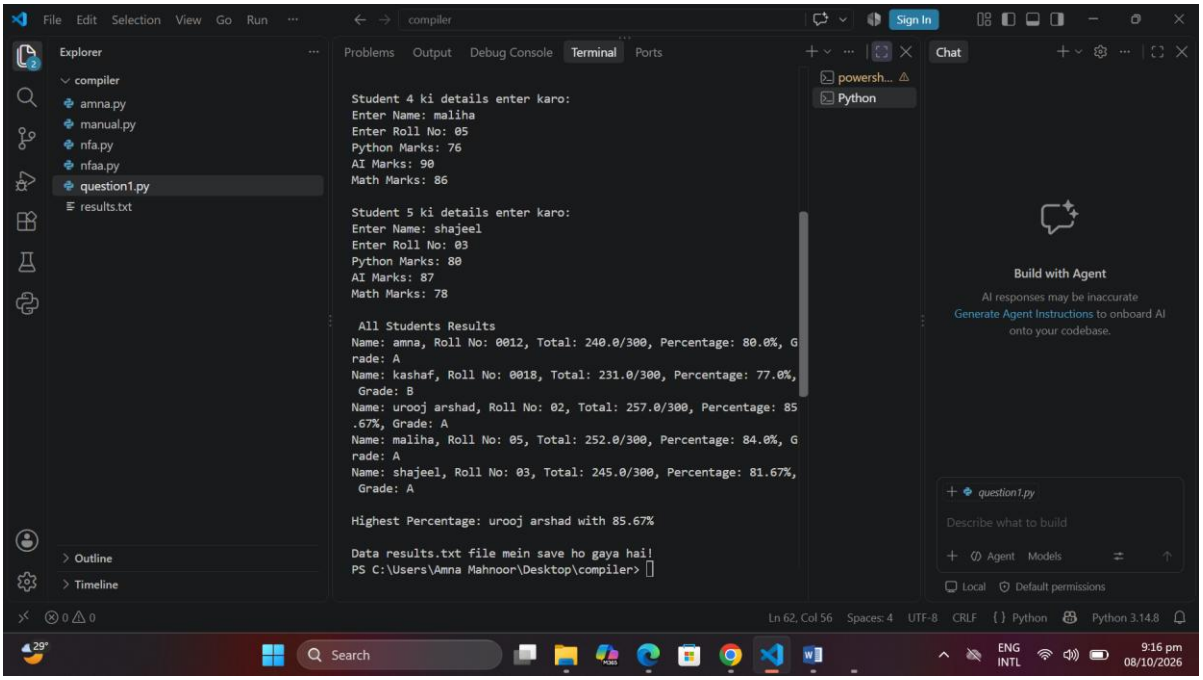
Q#1) Write a Python program to create a simple student result management system.

Ans :




File Edit Selection View Go Run
compiler
Sign In
Explorer
compiler
amna.py
manual.py
nfa.py
nfaa.py
question1.py
results.txt
nfa.py
manual.py
amna.py
question1.py
Chat
"total": total,
"percentage": percentage,
"grade": grade
}
students_list.append(student_data)
print()
print(" All Students Results ")
file_text = "Student Results\n\n"
highest_per = 0
top_student_name = ""
for s in students_list:
line = "Name: " + s["name"] + ", Roll No: " + s["roll"] + ",
print(line)
file_text = file_text + line + "\n"
if s["percentage"] > highest_per:
highest_per = s["percentage"]
top_student_name = s["name"]
top_line = "\nHighest Percentage: " + top_student_name + " with
print(top_line)
file_text = file_text + top_line
f = open("results.txt", "w")
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
question1.py
Describe what to build
Agent
Models
Local
Default permissions
Outline
Timeline
Ln 9, Col 5
Spaces: 4
UTF-8
CRLF
Python
Python 3.14.8
31°C
Mostly clear
Search
ENG
INTL
9:13 pm
08/10/2026

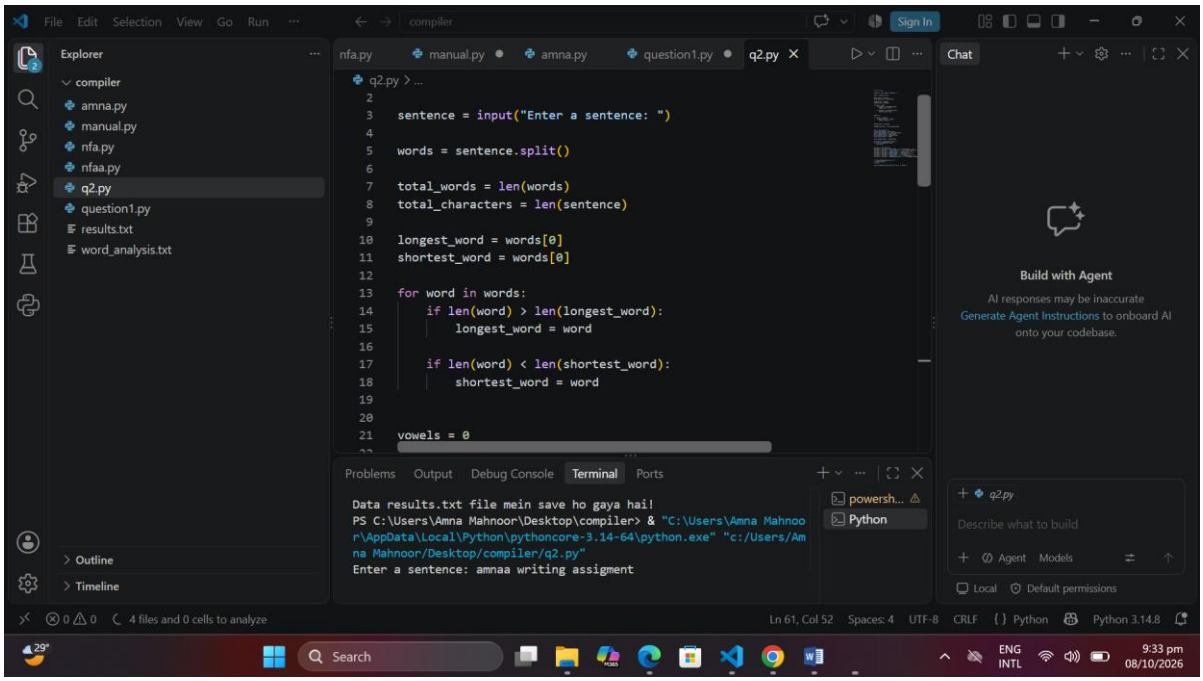

File Edit Selection View Go Run
compiler
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nfaa.py
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results.txt
nfa.py
manual.py
amna.py
question1.py
Chat
top_line = "\nHighest Percentage: " + top_student_name + " with
print(top_line)
file_text = file_text + top_line
f = open("results.txt", "w")
f.write(file_text)
f.close()
print("\nData results.txt file mein save ho gaya hai!")
Problems
Output
Debug Console
Terminal
Ports
powersh...
Python
Student 1 ki details enter karo:
Enter Name: amna
Enter Roll No: 0012
Python Marks: 80
AI Marks: 80
Math Marks: 80
Student 2 ki details enter karo:
Enter Name: kashaf
Enter Roll No: 0018
Python Marks: 81
AI Marks: 75
Math Marks: 75
Student 3 ki details enter karo:
Enter Name: urooj arshad
Enter Roll No: 02
Build with Agent
AI responses may be inaccurate
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question1.py
Describe what to build
Agent
Models
Local
Default permissions
Outline
Timeline
Ln 62, Col 56
Spaces: 4
UTF-8
CRLF
Python
Python 3.14.8
29°C
Mostly clear
Search
ENG
INTL
9:15 pm
08/10/2026

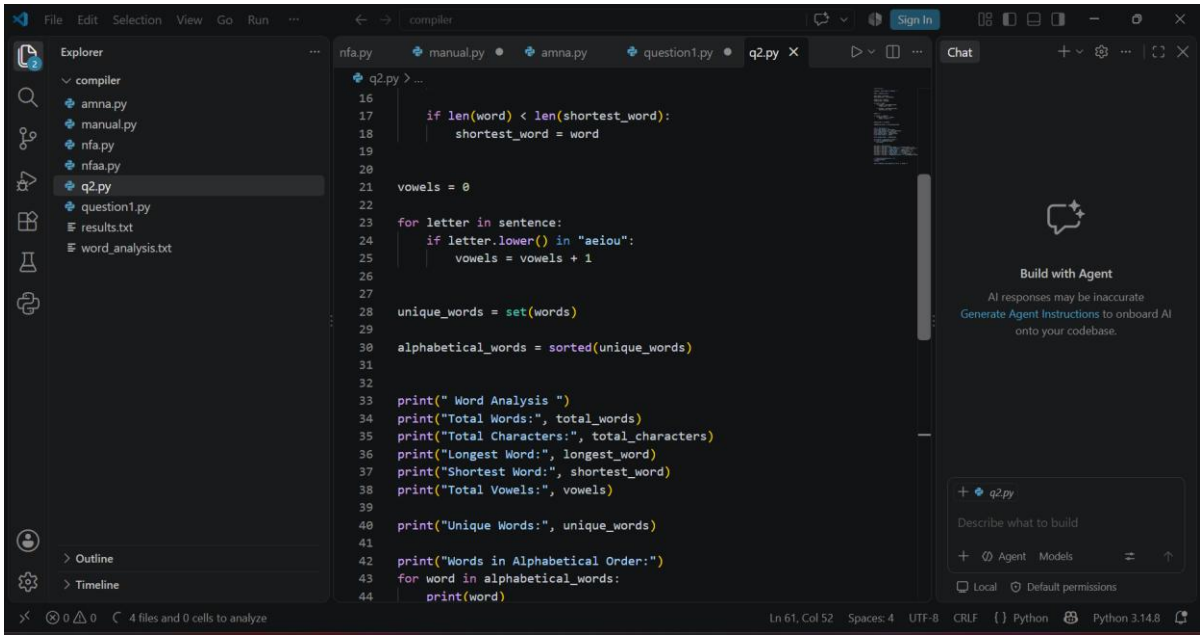


Q#2) Ask the user to enter a sentence.

The program should:

Ans:






```
    if len(word) < len(shortest_word):
        shortest_word = word


vowels = 0

for letter in sentence:
    if letter.lower() in "aeiou":
        vowels = vowels + 1

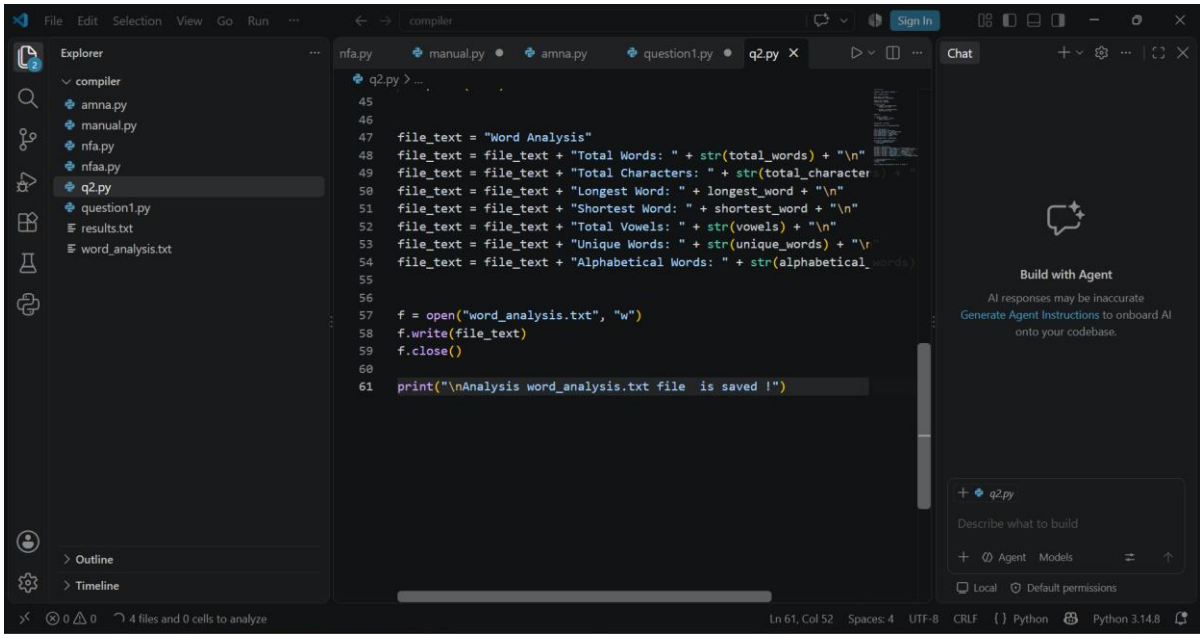

unique_words = set(words)

alphabetical_words = sorted(unique_words)


print(" Word Analysis ")
print("Total Words:", total_words)
print("Total Characters:", total_characters)
print("Longest Word:", longest_word)
print("Shortest Word:", shortest_word)
print("Total Vowels:", vowels)

print("Unique Words:", unique_words)

print("Words in Alphabetical Order:")
for word in alphabetical_words:
    print(word)
```

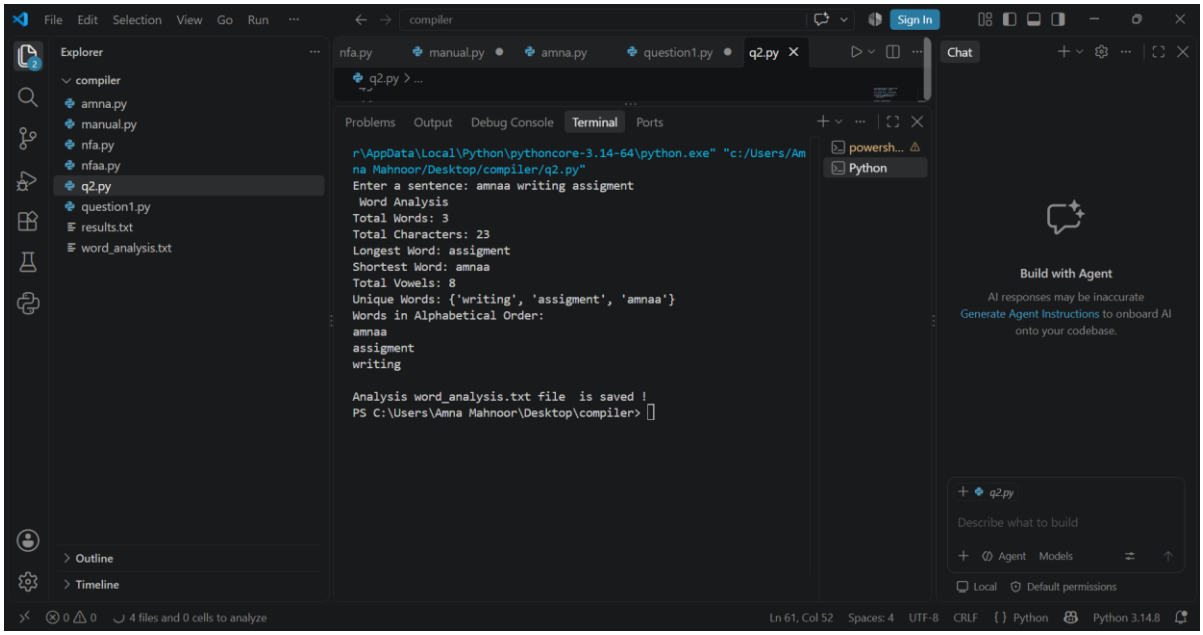



```


file_text = "Word Analysis"
file_text = file_text + "Total Words: " + str(total_words) + "\n"
file_text = file_text + "Total Characters: " + str(total_character
file_text = file_text + "Longest Word: " + longest_word + "\n"
file_text = file_text + "Shortest Word: " + shortest_word + "\n"
file_text = file_text + "Total Vowels: " + str(vowels) + "\n"
file_text = file_text + "Unique Words: " + str(unique_words) + "\r
file_text = file_text + "Alphabetical Words: " + str(alphabetical_

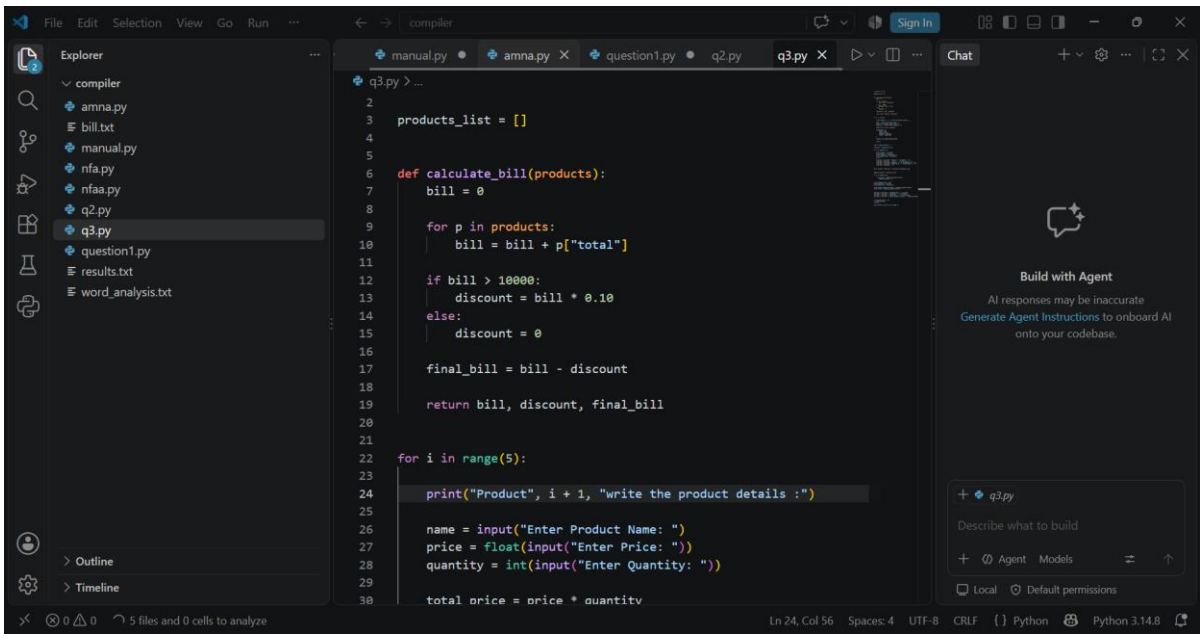

f = open("word_analysis.txt", "w")
f.write(file_text)
f.close()

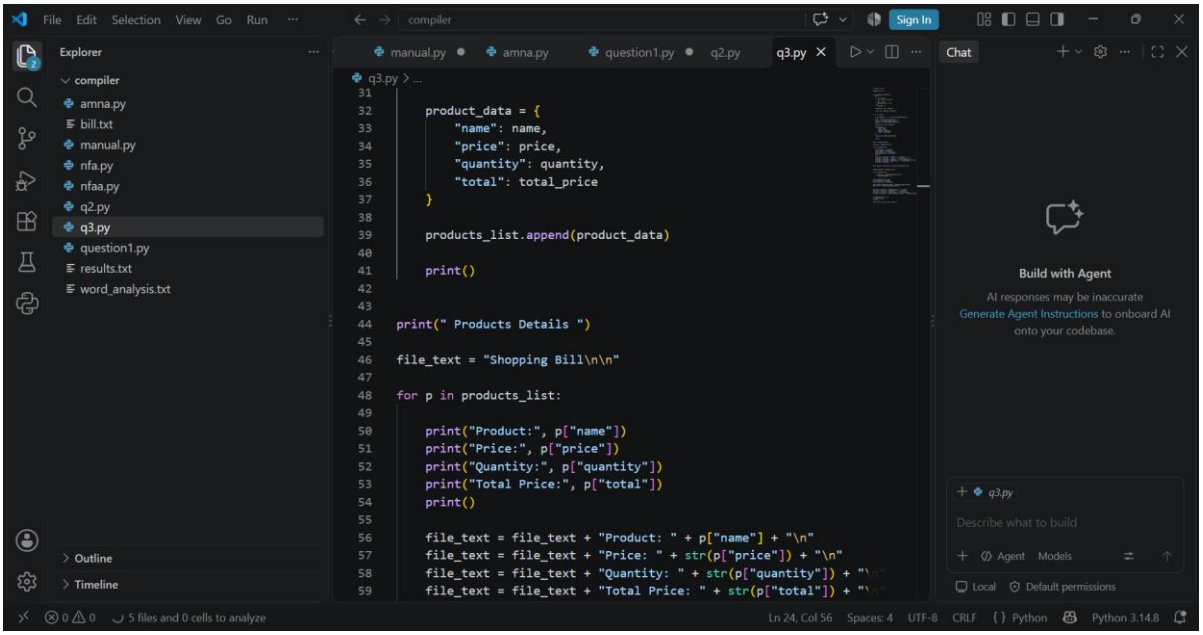
print("\nAnalysis word_analysis.txt file  is saved !")
```



Q#3) Create a simple shopping program.

Ans:





```
    product_data = {
        "name": name,
        "price": price,
        "quantity": quantity,
        "total": total_price
    }

    products_list.append(product_data)

    print()

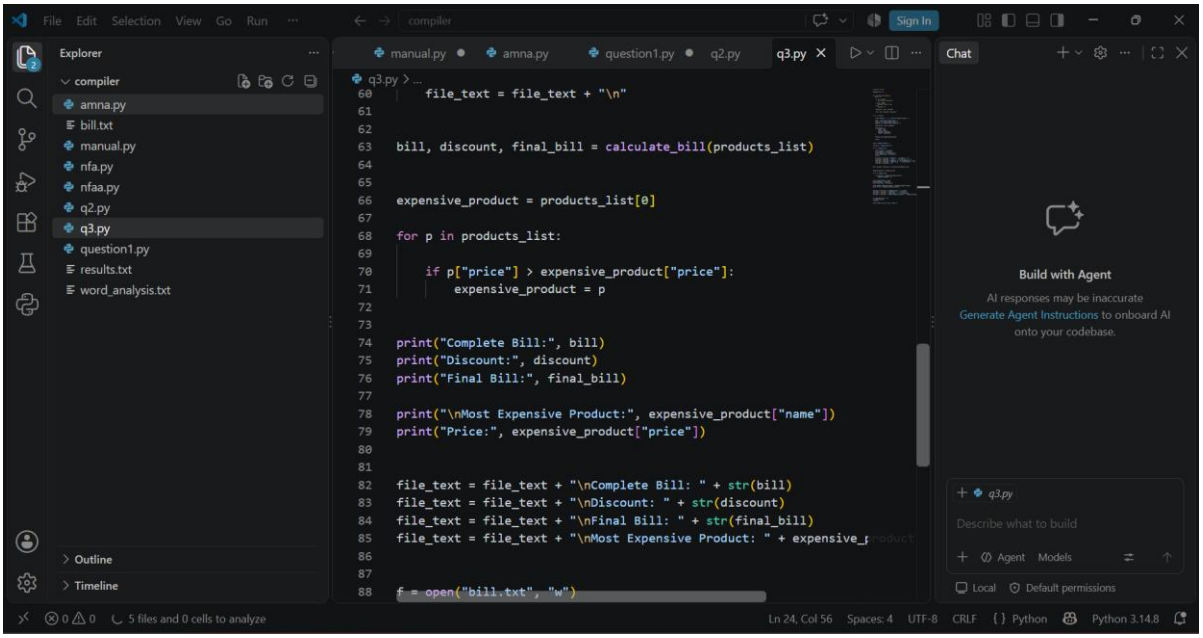

print(" Products Details ")

file_text = "Shopping Bill\n\n"

for p in products_list:

    print("Product:", p["name"])
    print("Price:", p["price"])
    print("Quantity:", p["quantity"])
    print("Total Price:", p["total"])
    print()

    file_text = file_text + "Product: " + p["name"] + "\n"
    file_text = file_text + "Price: " + str(p["price"]) + "\n"
    file_text = file_text + "Quantity: " + str(p["quantity"]) + "\
    file_text = file_text + "Total Price: " + str(p["total"]) + "\
```



```
    file_text = file_text + "\n"


bill, discount, final_bill = calculate_bill(products_list)


expensive_product = products_list[0]

for p in products_list:

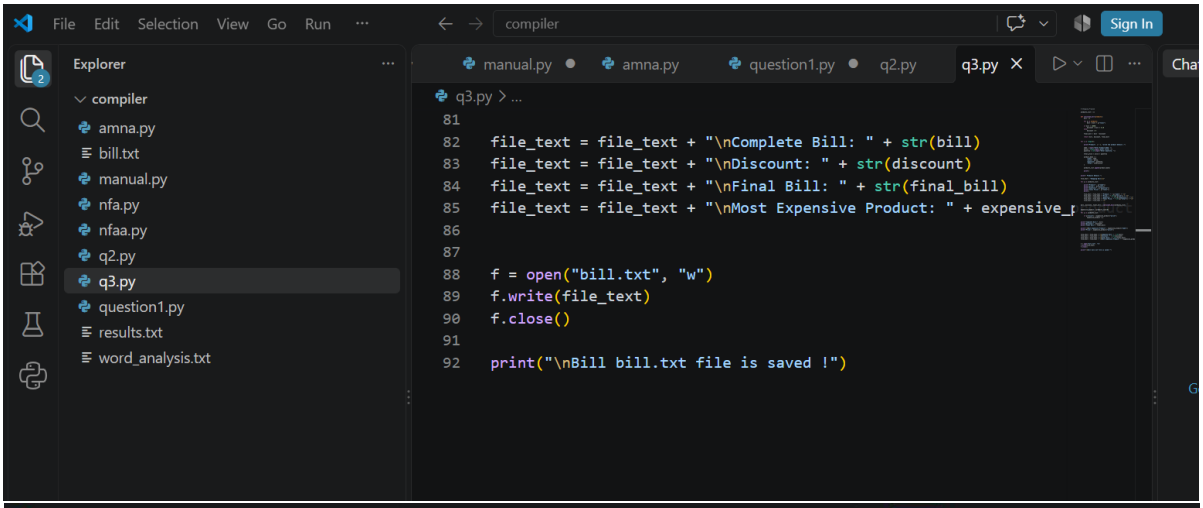
    if p["price"] > expensive_product["price"]:
        expensive_product = p

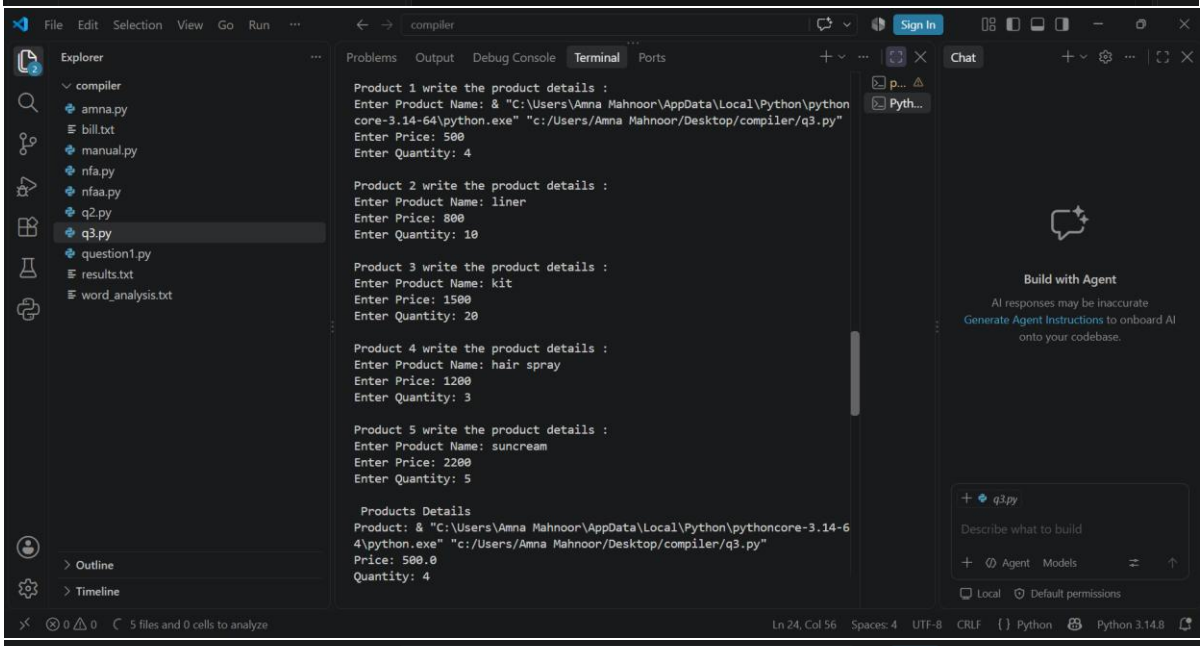

print("Complete Bill:", bill)
print("Discount:", discount)
print("Final Bill:", final_bill)

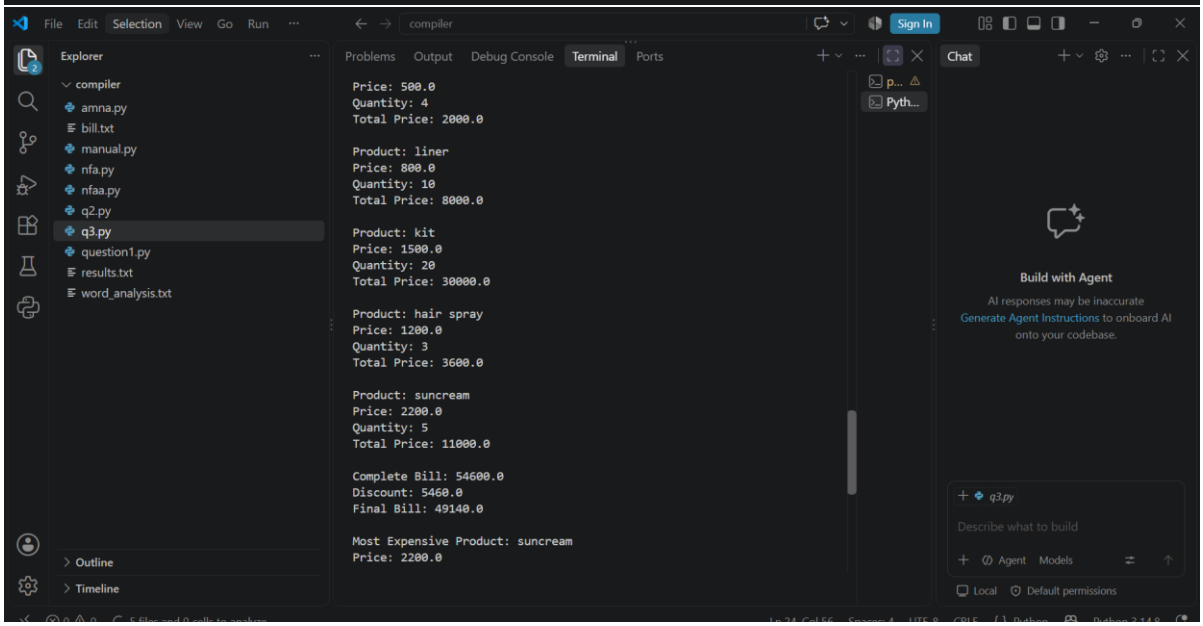
print("\nMost Expensive Product:", expensive_product["name"])
print("Price:", expensive_product["price"])


file_text = file_text + "\nComplete Bill: " + str(bill)
file_text = file_text + "\nDiscount: " + str(discount)
file_text = file_text + "\nFinal Bill: " + str(final_bill)
file_text = file_text + "\nMost Expensive Product: " + expensive_p

f = open("bill.txt", "w")
```

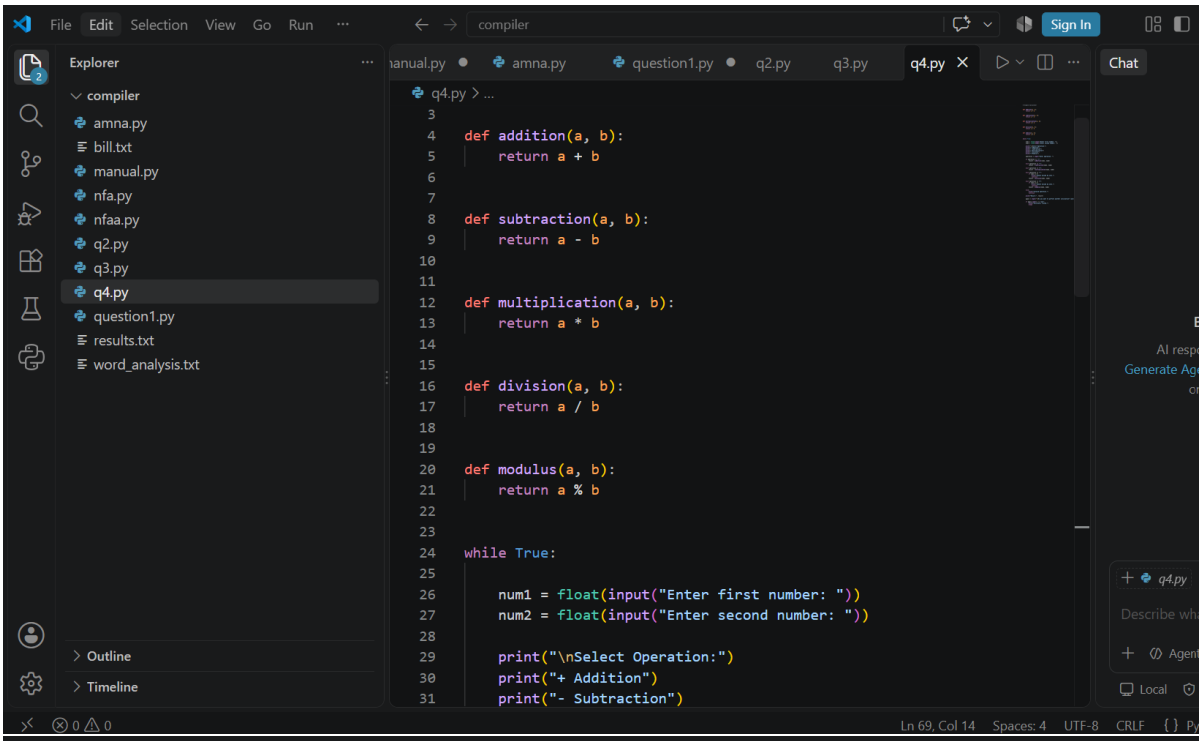

File Edit Selection View Go Run
compiler
Sign In
Explorer
compiler
amna.py
bill.txt
manual.py
nfa.py
nfaa.py
q2.py
q3.py
question1.py
results.txt
word_analysis.txt
manual.py
amna.py
question1.py
q2.py
q3.py
q3.py > ...
81
82 file_text = file_text + "\nComplete Bill: " + str(bill)
83 file_text = file_text + "\nDiscount: " + str(discount)
84 file_text = file_text + "\nFinal Bill: " + str(final_bill)
85 file_text = file_text + "\nMost Expensive Product: " + expensive_p
86
87
88 f = open("bill.txt", "w")
89 f.write(file_text)
90 f.close()
91
92 print("\nBill bill.txt file is saved !")


Problems Output Debug Console Terminal Ports
Chat
Product 1 write the product details :
Enter Product Name: & "C:\Users\Amna Mahnoor\AppData\Local\Python\pythoncore-3.14-64\python.exe" "c:/Users/Amna Mahnoor/Desktop/compiler/q3.py"
Enter Price: 500
Enter Quantity: 4
Product 2 write the product details :
Enter Product Name: liner
Enter Price: 800
Enter Quantity: 10
Product 3 write the product details :
Enter Product Name: kit
Enter Price: 1500
Enter Quantity: 20
Product 4 write the product details :
Enter Product Name: hair spray
Enter Price: 1200
Enter Quantity: 3
Product 5 write the product details :
Enter Product Name: suncream
Enter Price: 2200
Enter Quantity: 5
Products Details
Product: & "C:\Users\Amna Mahnoor\AppData\Local\Python\pythoncore-3.14-64\python.exe" "c:/Users/Amna Mahnoor/Desktop/compiler/q3.py"
Price: 500.0
Quantity: 4
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Agent Models
Local Default permissions
Outline
Timeline
Ln 24, Col 56 Spaces: 4 UTF-8 CRLF Python Python 3.14.8


Problems Output Debug Console Terminal Ports
Chat
Price: 500.0
Quantity: 4
Total Price: 2000.0
Product: liner
Price: 800.0
Quantity: 10
Total Price: 8000.0
Product: kit
Price: 1500.0
Quantity: 20
Total Price: 30000.0
Product: hair spray
Price: 1200.0
Quantity: 3
Total Price: 3600.0
Product: suncream
Price: 2200.0
Quantity: 5
Total Price: 11000.0
Complete Bill: 54600.0
Discount: 5460.0
Final Bill: 49140.0
Most Expensive Product: suncream
Price: 2200.0
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Agent Models
Local Default permissions
Outline
Timeline

Q#4) Write a Python program that works as a calculator. The program should:

Ans :



```
def addition(a, b):
    return a + b


def subtraction(a, b):
    return a - b


def multiplication(a, b):
    return a * b

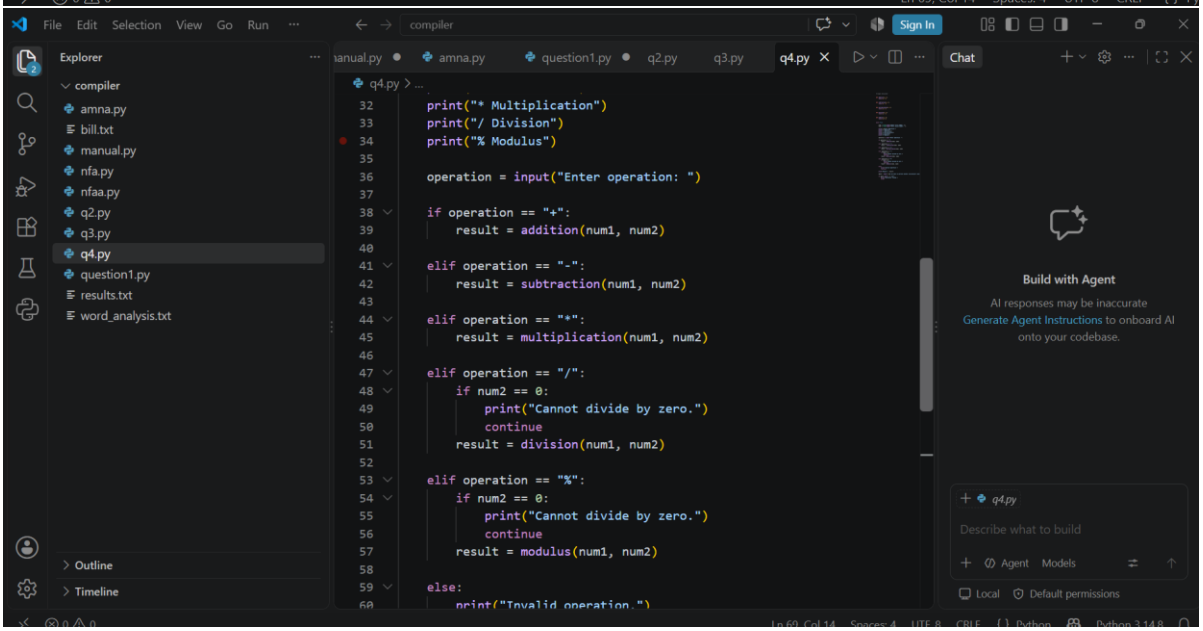

def division(a, b):
    return a / b


def modulus(a, b):
    return a % b


while True:

    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))

    print("\nSelect Operation:")
    print("+ Addition")
    print("- Subtraction")
```



```
    print("* Multiplication")
    print("/ Division")
    print("% Modulus")

    operation = input("Enter operation: ")

    if operation == "+":
        result = addition(num1, num2)

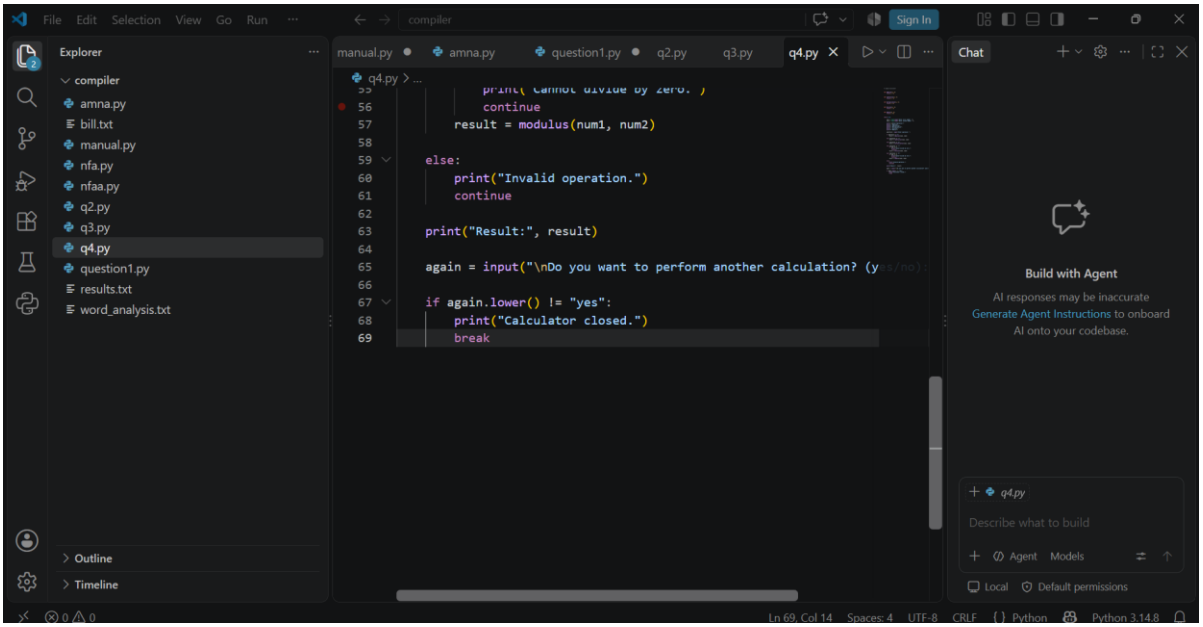
    elif operation == "-":
        result = subtraction(num1, num2)

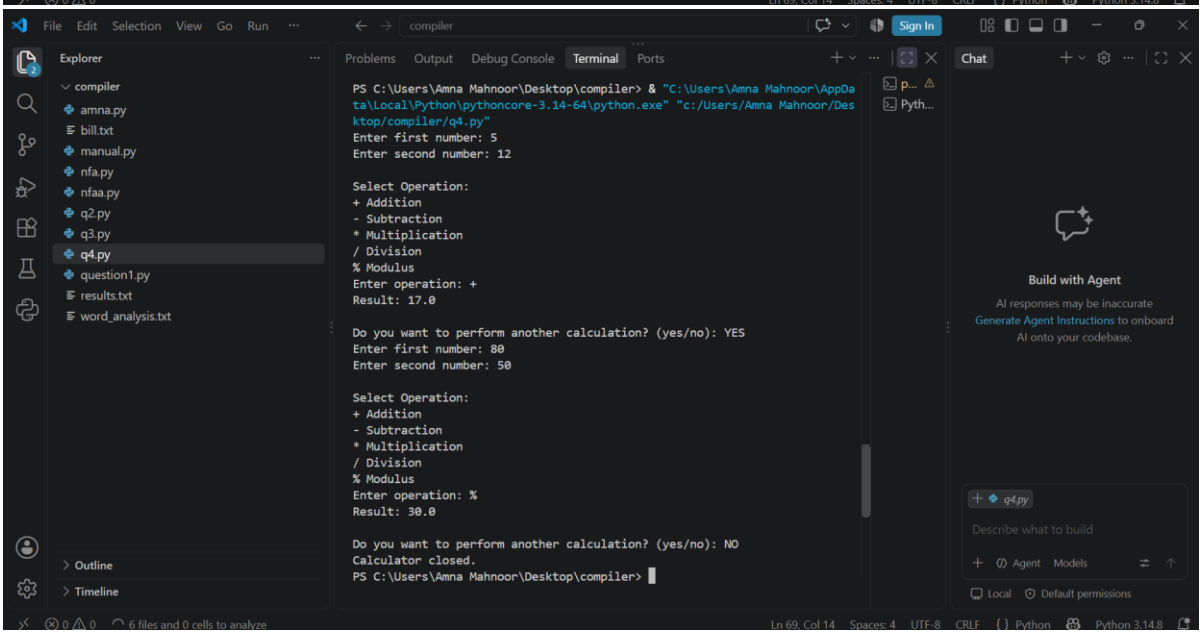
    elif operation == "*":
        result = multiplication(num1, num2)

    elif operation == "/":
        if num2 == 0:
            print("Cannot divide by zero.")
            continue
        result = division(num1, num2)

    elif operation == "%":
        if num2 == 0:
            print("Cannot divide by zero.")
            continue
        result = modulus(num1, num2)

    else:
```

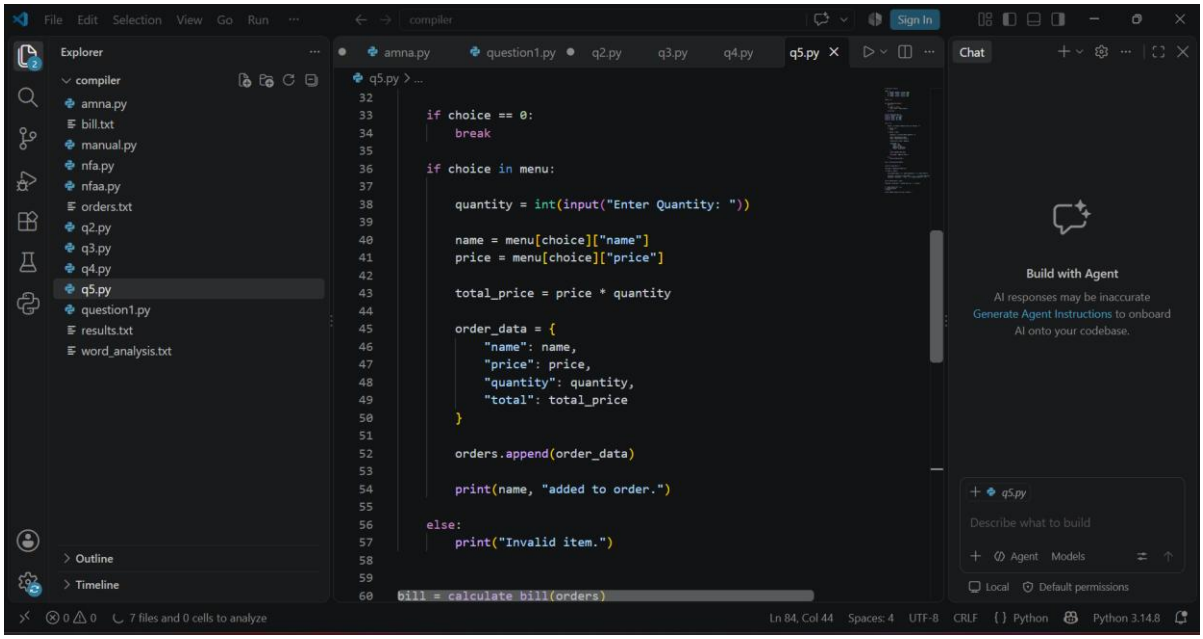




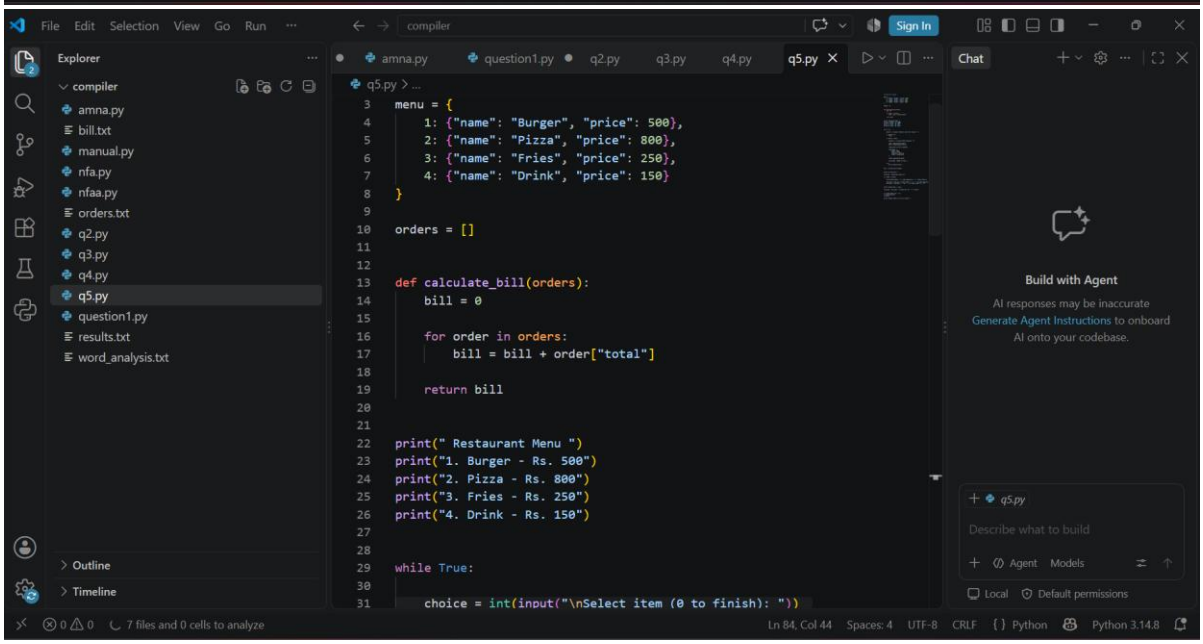
Q#5) Create a program for a small restaurant.

Display a menu such as:

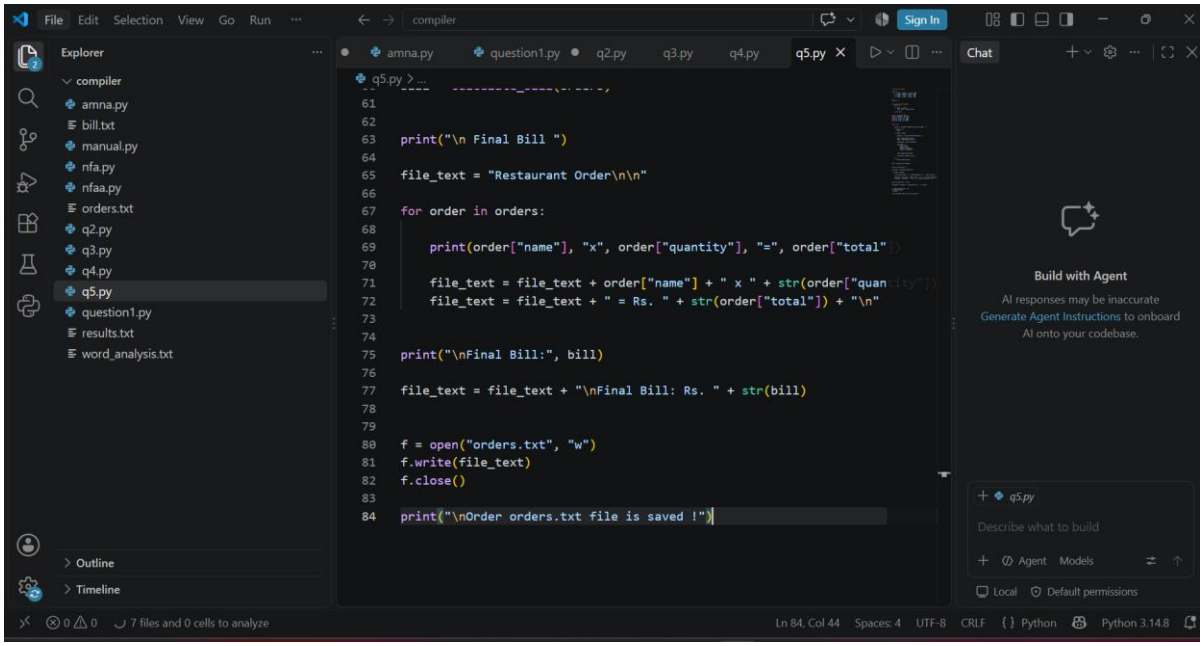
Ans :




```
32
33      if choice == 0:
34          break
35
36      if choice in menu:
37
38          quantity = int(input("Enter Quantity: "))
39
40          name = menu[choice]["name"]
41          price = menu[choice]["price"]
42
43          total_price = price * quantity
44
45          order_data = {
46              "name": name,
47              "price": price,
48              "quantity": quantity,
49              "total": total_price
50          }
51
52          orders.append(order_data)
53
54          print(name, "added to order.")
55
56      else:
57          print("Invalid item.")
58
59
60  bill = calculate_bill(orders)
```




```
3   menu = {
4       1: {"name": "Burger", "price": 500},
5       2: {"name": "Pizza", "price": 800},
6       3: {"name": "Fries", "price": 250},
7       4: {"name": "Drink", "price": 150}
8   }
9
10  orders = []
11
12
13  def calculate_bill(orders):
14      bill = 0
15
16      for order in orders:
17          bill = bill + order["total"]
18
19      return bill
20
21
22  print(" Restaurant Menu ")
23  print("1. Burger - Rs. 500")
24  print("2. Pizza - Rs. 800")
25  print("3. Fries - Rs. 250")
26  print("4. Drink - Rs. 150")
27
28
29  while True:
30
31      choice = int(input("\nSelect item (0 to finish): "))
```



```
print("\n Final Bill ")

file_text = "Restaurant Order\n\n"

for order in orders:

    print(order["name"], "x", order["quantity"], "=", order["total"])

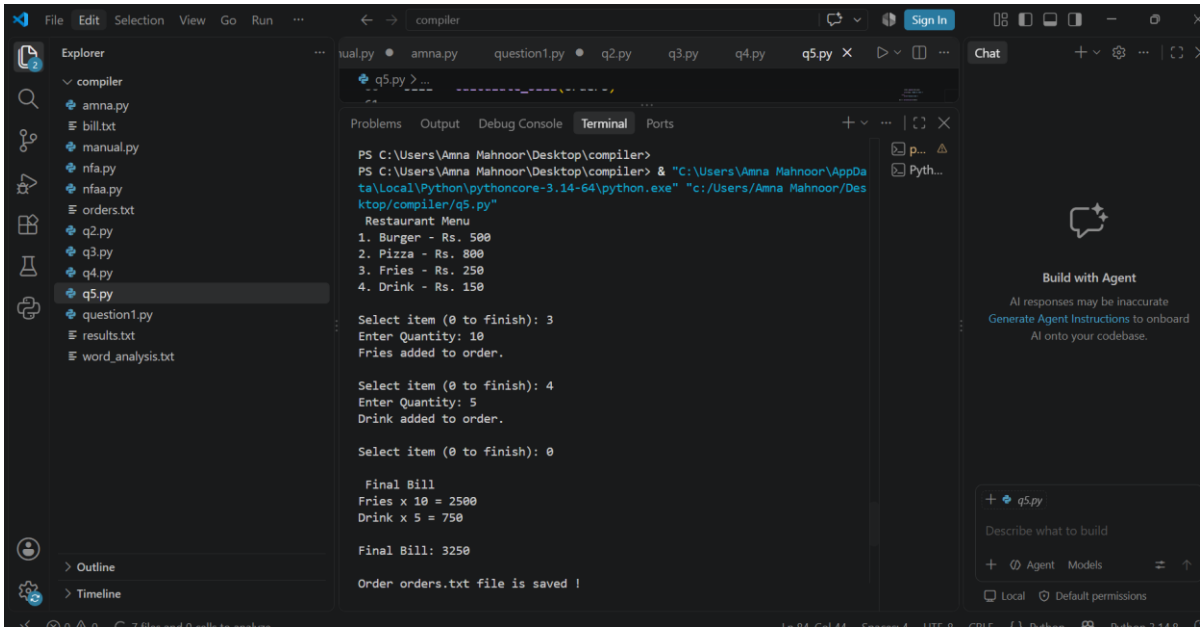
    file_text = file_text + order["name"] + " x " + str(order["quantity"])
    file_text = file_text + " = Rs. " + str(order["total"]) + "\n"


print("\nFinal Bill:", bill)

file_text = file_text + "\nFinal Bill: Rs. " + str(bill)


f = open("orders.txt", "w")
f.write(file_text)
f.close()

print("\nOrder orders.txt file is saved !")
```



```
PS C:\Users\Amna Mahnoor\Desktop\compiler>
PS C:\Users\Amna Mahnoor\Desktop\compiler> & "C:\Users\Amna Mahnoor\AppData\Local\Python\pythoncore-3.14-64\python.exe" "c:/Users/Amna Mahnoor/Desktop/compiler/q5.py"
 Restaurant Menu
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150

Select item (0 to finish): 3
Enter Quantity: 10
Fries added to order.

Select item (0 to finish): 4
Enter Quantity: 5
Drink added to order.

Select item (0 to finish): 0

 Final Bill
Fries x 10 = 2500
Drink x 5 = 750

Final Bill: 3250

Order orders.txt file is saved !
```